# WEAK DIMENSION DOES NOT INCREASE ALONG A WEAKLY ETALE RING MAP

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The counted proof span displays the flat resolution 0 -> K -> F_{d-1} -> ... -> F_0 -> N -> 0, delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

*Allowed standard prerequisites (editorial).*

- The key statement relating flatness over A and over B along a weakly etale map. invoked at proof lines 30238 and 30243. Upgrades A-flatness to B-flatness in both cases.
- The criterion identifying flatness of the relevant syzygy with a bound on weak dimension. invoked at proof lines 30241-30242. Produces the A-flat syzygy K at position d.
- The definition of tor dimension via flat resolutions. used at proof lines 30244-30246. Converts the existence of a flat resolution of length d into the conclusion.

*Source reference.* Original lemma 30202: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30202–30202. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 16883: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $A \to B$ be a weakly étale ring map. If $A$ has weak dimension at most $d$, then so does $B$.*

*Proof.* Let $N$ be a $B$-module. If $d = 0$, then $N$ is flat as an $A$-module, hence flat as a $B$-module by Lemma 30202. Assume $d > 0$. Choose a resolution $F_\bullet \to N$ by free $B$-modules. Our assumption implies that $K = \mathrm{Im}(F_d \to F_{d-1})$ is $A$-flat, see Lemma 16883. Hence it is $B$-flat by Lemma 30202. Thus $0 \to K \to F_{d-1} \to \ldots \to$

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$F_0 \to N \to 0$ is a flat resolution of length $d$ and we see that $N$ has tor dimension at most $d$. $\square$